

CIS 421 Database Management Systems Final Project

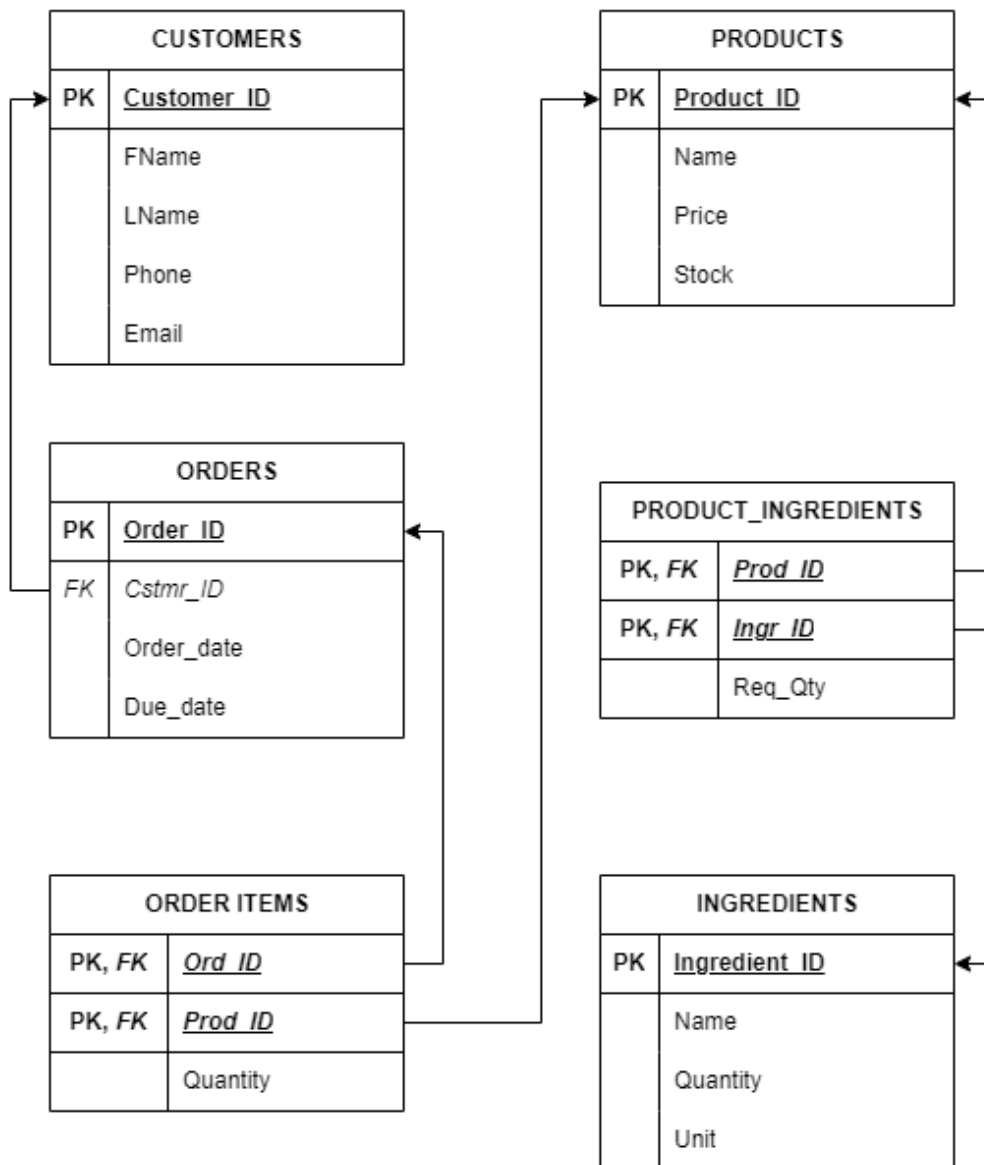
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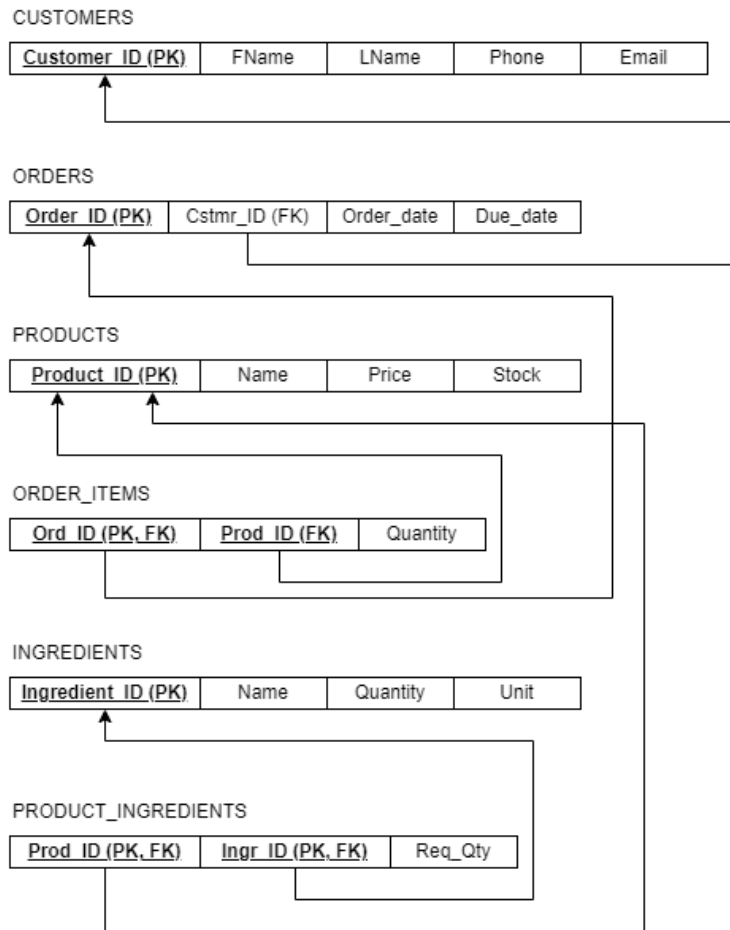
Relational Model

- **Objective:** Convert your conceptual model into a **Relational Model** in **3NF**.
- **Content:** Illustrate the resulting relations (tables) and their relationships, confirming that they meet 3NF requirements for normalization.

Relation Schema



Relational Model



Normal Forms

1NF

Every attribute in a relation must contain atomic values (no repeating groups or multivalues) and every row must be identifiable by its primary key to be in 1NF.

- **✓ CUSTOMERS**
Primary Key: (Customer ID)
Foreign Key(s): N/A
 - There are no repeating groups or multivalues.
 - All attributes can be determined by Customer ID.

- **✓ ORDERS**
Primary Key: (Order_ID)
Foreign Key(s): Cstmr_ID (FK: Customers.Customer_ID)
 - There are no repeating groups or multivalues.
 - All attributes can be determined by **Order_ID**.
- **✓ PRODUCTS**
Primary Key: (Product_ID)
Foreign Key(s): N/A
 - There are no repeating groups or multivalues.
 - All attributes can be determined by **Product_ID**.
- **✓ ORDER_ITEMS**
Primary Key: (Ord_ID, Prod_ID)
Foreign Key(s): Prod_ID (FK: Products.Product_ID)
 - There are no repeating groups or multivalues.
 - All attributes can be determined by (**Ord_ID, Prod_ID**).
- **✓ INGREDIENTS**
Primary Key: (Ingredient_ID)
Foreign Key(s): N/A
 - There are no repeating groups or multivalues.
 - All attributes can be determined by **Ingredient_ID**.
- **✓ PRODUCT_INGREDIENTS**
Primary Key: (Prod_ID, Ingr_ID)
Foreign Key(s): Prod_ID (FK: Products.Product_ID), Ingr_ID (FK: Ingredients.Ingredient_ID)
 - There are no repeating groups or multivalues.
 - All attributes can be determined by (**Prod_ID, Ingr_ID**).

All attributes in all relations contain atomic values with no repeating groups. Every record in every relation can also be determined by its primary key. Therefore, all relations are in 1NF.

2NF

All relations are in 1NF as stated above. All non-key attributes must be fully dependent on the entire primary key with no partial dependencies on a part of a composite key to be in 2NF.

- **✓ CUSTOMERS**
Primary Key: (Customer_ID)
Foreign Key(s): N/A
 - All non-key attributes depend entirely on **Customer_ID**.

- **✓ ORDERS**
*Primary Key: (**Order_ID**)*
Foreign Key(s): Cstmr_ID (FK: Customers.Customer_ID)
 - All non-key attributes depend entirely on **Order_ID**.
 - FK: Cstmr_ID references Customers.Customer_ID, and it is directly related to Order_ID.
- **✓ PRODUCTS**
*Primary Key: (**Product_ID**)*
Foreign Key(s): N/A
 - All non-key attributes depend entirely on **Product_ID**.
- **✓ ORDER_ITEMS**
*Primary Key: (**Ord_ID, Prod_ID**)*
*Foreign Key(s): **Prod_ID** (FK: Products.Product_ID)*
 - The non-key attribute (Quantity) depends entirely on the composite primary key. It cannot be determined by **Ord_ID** or **Prod_ID** alone.
 - There are no partial dependencies.
- **✓ INGREDIENTS**
*Primary Key: (**Ingredient_ID**)*
Foreign Key(s): N/A
 - All non-key attributes depend entirely on **Ingredient_ID**.
- **✓ PRODUCT_INGREDIENTS**
*Primary Key: (**Prod_ID, Ingr_ID**)*
*Foreign Key(s): **Prod_ID** (FK: Products.Product_ID), **Ingr_ID** (FK: Ingredients.Ingredient_ID)*
 - The non-key attribute (Req_Qty) depends entirely on the composite primary key. It cannot be determined by **Prod_ID** or **Ingr_ID** alone.
 - There are no partial dependencies.

All relations in the database depend entirely on their primary keys whether they are non-composite or composite. All non-key attributes in all of the relations depend on their entire primary keys. Therefore, there are no partial dependencies in the database and all relations are in 2NF.

3NF

All relations are in 2NF as stated above. All non-key attributes must be fully dependent on the entire primary key and cannot be transitively dependent on other non-key attributes to be in 3NF.

- **✓ CUSTOMERS**
Primary Key: (Customer_ID)
Foreign Key(s): N/A
 - **Customer_ID** uniquely identifies each Customer.
 - All other attributes depend on **Customer_ID**, not on one another.
 - There are no non-key attributes that depend on other non-key attributes.
- **✓ ORDERS**
Primary Key: (Order_ID)
Foreign Key(s): Cstmr_ID (FK: Customers.Customer_ID)
 - **Order_ID** uniquely identifies each Order.
 - The foreign key *Cstmr_ID* does not create a transitive dependency because it references the primary key in the Customers table.
 - All other attributes depend on **Order_ID**, not on one another.
 - There are no non-key attributes that depend on other non-key attributes.
- **✓ PRODUCTS**
Primary Key: (Product_ID)
Foreign Key(s): N/A
 - **Product_ID** uniquely identifies each Product.
 - All other attributes depend on **Product_ID**, not on one another.
 - There are no non-key attributes that depend on other non-key attributes.
- **✓ ORDER_ITEMS**
Primary Key: (Ord_Id, Prod_ID)
Foreign Key(s): Ord_ID (FK: Orders.Order_ID), Prod_ID (FK: Products.Product_ID)
 - (**Ord_Id**, **Prod_ID**) uniquely identifies each Order_Item.
 - The foreign keys *Ord_ID* and *Prod_ID* do not create transitive dependencies because they reference primary keys in the Orders and Products tables.
 - All other attributes depend entirely on (**Ord_Id**, **Prod_ID**), not on one another.
 - There are no non-key attributes that depend on other non-key attributes.
- **✓ INGREDIENTS**
Primary Key: (Ingredient_ID)
Foreign Key(s): N/A
 - **Ingredient_ID** uniquely identifies each Ingredient.
 - All other attributes depend on **Ingredient_ID**, not on one another.
 - There are no non-key attributes that depend on other non-key attributes.
- **✓ PRODUCT_INGREDIENTS**
Primary Key: (Prod_ID, Ingr_ID)
Foreign Key(s): Prod_ID (FK: Products.Product_ID), Ingr_ID (FK: Ingredients.Ingredient_ID)

- (**Prod_ID, Ingr_ID**) uniquely identifies each Product_Ingredient.
- The foreign keys **Prod_ID** and **Ingr_ID** do not create transitive dependencies because they reference primary keys in the Products and Ingredients tables.
- All other attributes depend entirely on (**Prod_ID, Ingr_ID**), not on one another.
- There are no non-key attributes that depend on other non-key attributes.

All relations in the database depend entirely on their primary keys whether they are non-composite or composite. There are also no non-key attributes that depend on other non-key attributes. All non-key attributes depend entirely on their primary keys and with zero transitive dependencies. Therefore, all relations in the database are in 3NF.

3NF Tables & Relationships

